26900234 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFSQ304 - SPICES AND OLEORESINS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Recognize the importance of preservation in maintaining food safety. [U] / [CO1]

   a). It helps delay spoilage and maintain food in a safer and acceptable condition

   b). It increases the risk of food deterioration

   c). It eliminates the need for hygienic food handling

   d). It reduces the storage life of food

2. Identify the compound associated with the characteristic warm spicy flavour of cumin. [K] / [CO2]

   a). Linalool b). Cuminaldehyde

   c). Eugenol d). Anethole

3. Recognize the major volatile oil constituent responsible for the characteristic aroma of clove [K] / [CO3]

a). Eugenol b). Gingerol c). Piperine d). Curcumin

4. Select the correct sequence for essential-oil manufacture by steam distillation. [U] / [CO4]

a). Grinding → steam distillation → condensation → separation → filtration

b). Condensation → grinding → fermentation → separation

c). Extraction → fermentation → drying → condensation

d). Grinding → solvent extraction → crystallization

5. Recognize the commercial spice obtained from cured and dried rhizomes. [K] / [CO3]

a). Nutmeg b). Turmeric

c). Fennel d). Clove

6. Recall how antioxidant spices can contribute to food preservation. [U] / [CO1]

a). They help delay oxidative deterioration and maintain food quality

b). They increase oxidation of fats

c). They accelerate rancidity

d). They reduce the storage stability of food

7. Examine the relationship between moisture content and spice trading. Which interpretation is most appropriate? [An] / [CO2]

a). Moisture can affect both product stability and the weight used for pricing

b). Moisture has no effect on traded weight once the spice is graded

c). Higher moisture always indicates better spice quality

d). Moisture determination is mainly used to estimate volatile-oil composition

8. Recognize the operation used to separate fenugreek seeds from harvested plants. [K] / [CO3]

a). Threshing

b). Peeling

c). Blanching

d). Fermentation

9. Recall the main purpose of solvent extraction in the manufacture of spice extracts. [K] / [CO4]

a). Removal of moisture only

b). Removal of proteins

c). Extraction of desired flavour and aroma compounds

d). Increasing fibre content

10. Select the appropriate stage for harvesting allspice berries for spice production. [K] / [CO3]

a). Mature but green stage b). Fully overripe stage

c). Completely rotten stage d). After seed germination

11. Recognize the major volatile oil constituent responsible for the characteristic aroma of cumin. [K] / [CO3]

a). Eugenol b). Gingerol c). Piperine d).Cuminaldehyde

12. Identify the operation that removes the outer papery skin of garlic: [U] / [CO3]

a). Grading b). Peeling c). Milling d). Sieving

13. Ginger oleoresin is obtained mainly by [K] / [CO3]

a). Fermentation b). Solvent extraction c). Freezing d). Salting

14. Recognize the major operation performed after drying in conventional turmeric processing to improve the external appearance of the cured rhizomes. [K] / [CO3]

a). Polishing b). Threshing

c). Retting d). Winnowing

15. Identify a common method used for obtaining cinnamon essential oil. [K] / [CO3]

a). Filtration only b). Crystallization

c). Sedimentation d). Steam or hydro-distillation

16. A bark spice is prepared into quills rather than being marketed primarily as an intact seed or fruit. The spice is ___. [A] / [CO3]

a). Cinnamon b). Coriander

c). Cumin d). Cardamom

17. A processor wants to produce black pepper rather than white pepper. The outer covering of the berries should be ___. [U] / [CO3]

a). Removed before drying b). Retained during drying

c). Removed by prolonged retting d). Dissolved by acid treatment

18. The component mainly responsible for the characteristic pungent odour of asafoetida. [K] / [CO3]

a). Sulphur-containing compounds b). Anthocyanins

c). Carotenoids d). Tannins

19. Identify the operation in which harvested spice material is separated from unwanted light material using air movement. [K] / [CO3]

a). Grading b). Winnowing

c). Curing d). Retting

20. Identify the quality attribute of chilli that is commonly expressed through ASTA colour value. [K] / [CO3]

a). Pungency b). Aroma

c). Colour strength d). Moisture

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Describe the meaning and significance of spices in food preparation. [U] / [CO1]

22. List the major steps involved in the processing and manufacture of mace spice. [K] / [CO3]

23. Identify the major by-products generated during oleoresin manufacture. [K] / [CO4]

24. Outline the major steps involved in the manufacture of spice oleoresins using supercritical CO_2. [K] / [CO4]

25. Detail on the chemical composition of black pepper. [K] / [CO2]

26. Recall the importance of curing in vanilla processing [K] / [CO3]

27. Explain the importance of harvesting cardamom capsules at the appropriate stage. [U] / [CO3]

28. Explain the need for careful handling of saffron after harvesting. [U] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Define oleoresins and describe their chemical composition. Explain the role of volatile and non-volatile constituents in determining the characteristic properties of oleoresins. [U] / [CO4]

30. Explain the manufacture of two different value-added coriander products. [U] / [CO3]

31. Describe how spices contribute to the sensory quality and acceptability of foods through their effects on flavour, taste, aroma, texture and colour. [U] / [CO1]

32. Explain the dietary and medicinal uses of cinnamon, clove, fennel, fenugreek, nutmeg, mace and saffron. [U] / [CO2]

[10x2 = 20]